

# Seasonal Variation Mediates the Importance of Species Attributes in Plant–Pollinator Interactions

Jose B. Lanuza<sup>1,2,\*</sup>, Alfonso Allen-Perkins<sup>3</sup>, Will Glenny<sup>1</sup>, Anna Traveset<sup>2</sup>, Henriette F. Morgenroth<sup>1</sup>, Raymond Umazekabiri<sup>1</sup>, Marc Hoffmann<sup>1</sup>, Panagiotis Theodorou<sup>1,4</sup>, Robert J. Paxton<sup>1,4</sup>, Isabell Hensen<sup>1,5</sup>, Robert Rauschkolb<sup>1,6,7</sup>, Christine Römermann<sup>1,6,7</sup>, Oliver Schweiger<sup>1,8</sup>, and Tiffany M. Knight<sup>1</sup>

<sup>1</sup> German Centre for Integrative Biodiversity Research (iDiv) Halle–Jena–Leipzig, Puschstr. 4, 04103 Leipzig, Germany.

<sup>2</sup> Mediterranean Institute for Advanced Studies (IMEDEA, UIB-CSIC), Mallorca, Balearic Islands, Spain.

<sup>3</sup> Departamento de Ingeniería Eléctrica, Electrónica, Automática y Física Aplicada, ETSIDI, Universidad Politécnica de Madrid, Madrid, Spain.

<sup>4</sup> Institute for Biology, General Zoology, Martin Luther University Halle-Wittenberg, Halle (Saale), Germany.

<sup>5</sup> Institute of Biology/Geobotany and Botanical Garden, Martin Luther University Halle-Wittenberg, Halle (Saale), Germany.

<sup>6</sup> Institute of Biodiversity, Ecology and Evolution, Friedrich Schiller University Jena, Germany.

<sup>7</sup> Senckenberg Institute for Plant Form and Function (SIP), Jena, Germany.

<sup>8</sup> Department of Community Ecology, UFZ–Helmholtz Centre for Environmental Research, Halle, Germany.

---

**Running title:** Seasonal drivers of plant–pollinator interactions

**Article type:** Letter

**Keywords:** plant–pollinator interactions; phenology; abundance; trait matching; temporal resolution; ecological networks; visitation rate; random forest; pollination ecology; community ecology

**Abstract:** XXX words

**Main text:** XXXX words

**Text boxes:** None

**References:** XX

**Figures:** X

**Tables:** X

**Text boxes:** 0

## Statement of authorship

J.B.L. conceived the study. J.B.L., T.M.K., and O.S. designed the study. J.B.L. compiled the data, performed the analyses, and wrote the first draft of the manuscript. A.A.-P. contributed to analytical methods. All authors contributed to data collection and/or interpretation of the results, critically revised the manuscript, and approved the final version.

## Data accessibility statement

The data supporting the findings of this study are openly available in Zenodo at <https://doi.org/10.5281/zenodo.15183272>. The R code used to reproduce all analyses and figures is openly available in Zenodo at <https://doi.org/10.5281/zenodo.21243408>.

## Corresponding author

38 Jose B. Lanuza  
39 Mediterranean Institute for Advanced Studies (IMEDEA, UIB-CSIC)  
40 07190 Esporles, Mallorca, Balearic Islands, Spain  
41 E-mail: jose.lanuza@imedea.uib-csic.es  
42 Telephone: +34 XXX XXX XXX